

Scott Nguyen

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EDUCATION

Ph.D. in Mechanical and Aerospace Engineering UC San Diego San Diego State University (<i>Joint Doctoral Program</i>)	Spring 2030
Master of Science in Electrical Engineering Space Systems Engineering University of New Mexico	Summer 2026 GPA: 3.93/4.00
Master of Science in Aerospace Engineering University of Illinois Urbana-Champaign	Fall 2024
Bachelor of Science in Aerospace Engineering Iowa State University	Spring 2022

SKILLS

Programming: Python, MATLAB, C/C++, Ruby

GNC / Flight Software: cFS, PX4, MAVSDK, MAVLink, ROS2, EKF, UKF, MEKF, Sensor Fusion, Orbit Determination

Simulation / Tools: Gazebo, Simulink, Trick, NumPy, SciPy, Git, Linux, CI/CD

Applications: ANSYS HFSS, FEKO, TICRA, Blender

WORK EXPERIENCE

Guidance, Navigation & Controls Engineer Intern **May 2026 – August 2026**
Air Force Research Laboratory (AFRL), Eglin Air Force Base

- Implemented and validated a **Pose Unit Dual Quaternion Multiplicative Extended Kalman Filter (PUDQ-MEKF)** against a traditional **Extended Kalman Filter (EKF)** for nonlinear pose estimation.
- Performed **Monte Carlo** analysis using **Normalized Estimation Error Squared (NEES)**, state-error, and covariance metrics to evaluate estimator consistency.
- Designed a keyframe-based relative pose estimation framework and quantified filter consistency through covariance, error-state, and trajectory-level performance analysis.
- Integrated the estimator in **Gazebo** and verified closed-loop performance on **ROS2**-based TurtleBot and multirotor platforms.

Guidance, Navigation & Controls Engineer Intern **January 2026 – May 2026**
NASA Johnson Space Center (JSC)

- Built Guidance, Navigation, and Controls (GNC) flight software within NASA's **core Flight System (cFS)** for the **SPLICE** project.
- Supported **frame camera**, **event camera**, and **Inertial Measurement Unit (IMU)** applications for telemetry, packet routing, and payload integration.
- Created **PX4 + MAVSDK** simulations and **Trick**-based tools for autonomous guidance and controller evaluation.

Student Co-Op **September 2025 – January 2026**
MIT Lincoln Laboratory, Electronics for Contested Space Group

- Built an optical Space Domain Awareness (SDA) tool to propagate Two-Line Element (TLE)-based Resident Space Object (RSO) orbits and identify observable tracking windows.
- Modeled RSO detectability and tracking performance using photometric, kinematic, and sensor-noise models.
- Designed a radio-frequency communication architecture to support lost-satellite reacquisition.

Guidance, Navigation & Controls Engineer Intern **May 2025 – August 2025**
Blue Canyon Technologies

- Executed post-environment testing of **IMUs**, star trackers, reaction wheels, torque rods, and sun sensors.
- Diagnosed **Solar Array Drive Assembly (SADA)** timing latency and automated **COSMOS (Ruby)** validation.

Guidance, Navigation & Controls Engineer Intern **January 2025 – April 2025**
Blue Origin

- Designed a hybrid **Active Disturbance Rejection Control (ADRC) + Sliding Mode Control (SMC)** controller for robust tracking under uncertainty and actuator limits.
- Benchmarked the hybrid controller against **PID** and **LQR** using disturbance-injected simulations and performance metrics.
- Integrated flight software into **Simulink** through **C S-functions** for closed-loop testing.

Guidance, Navigation & Controls Engineer Intern **May 2024 – August 2024**
Varda Space Industries

- Performed gravity-model and **EKF vs. Unscented Kalman Filter (UKF)** trade studies for reentry navigation architecture selection.
- Implemented an **EKF** and Monte Carlo flight-safety analyses for state estimation, residual reduction, and delta-v planning.

Guidance, Navigation & Controls Engineer Intern **May 2023 – August 2023**
Space Dynamics Laboratory

- Implemented angles-only orbit determination algorithms using **UKF**, least-squares, and range-iteration methods for Space Situational Awareness (SSA).
- Performed Monte Carlo observability studies and implemented a **Lambert-based Initial Orbit Determination (IOD)** solver.